

2024 (NEP) (Freshers)

COMPUTER SCIENCE

Data Analytics

Time : 2½ Hours

Max. Marks : 60

instruction : Answer all the Sections.

SECTION – A

I. Answer any four questions. Each question carries two marks. (4x2=8)

1) Define the term Data Analytics.

Data analytics is the process of collecting, organizing, analyzing and interpreting raw data in order to discover useful insights, identify patterns, support decision making and solve real-world problems.

Example-A company analyzes customers' purchase data to know which products are sold most and plans stock based on it.

2) Name any four data visualization tools used.

Tableau

Tableau is a powerful data visualization tool used to create interactive dashboards and reports.

It helps analyze large datasets and present insights visually for better decision-making.

Power BI

Power BI is a business analytics and visualization tool from Microsoft used to build real-time dashboards and reports.

It easily integrates with Excel, databases, and cloud services to analyze and share data.

Google Charts

Google Charts is a free web-based tool that uses JavaScript to create interactive charts and graphs for websites.

It supports many chart types and can connect with Google Sheets or online data sources.

D3.js

D3.js (Data-Driven Documents) is a JavaScript library used for creating highly customized and interactive visualizations.

It uses HTML, SVG, and CSS to build complex and dynamic data graphics.

3) Explain the term Normal Distribution.

Normal Distribution is a continuous probability distribution in which data is symmetrically distributed around the mean, forming a bell-shaped curve.

In a normal distribution, most values lie close to the mean, and the probability decreases equally on both sides as values move away from the mean.

Example -Students' exam marks, heights of people, blood pressure readings, and measurement errors often follow a normal distribution.

4) Define the following events :

i) Mutually exclusive

ii) Equally likely.

i) Mutually Exclusive Events

Mutually exclusive events are those events which cannot occur at the same time.

If one event happens, the other cannot happen, i.e., they have no common outcomes.

Example: When tossing a coin, getting Head and Tail are mutually exclusive events.

ii) Equally Likely Events

Equally likely events are events that have the same chance of occurring.

Each event has an equal probability of happening.

Example: In a fair dice roll, the probability of getting 1, 2, 3, 4, 5, or 6 are equally likely.

5) What is power query ?

Power Query is a data connection and data transformation tool used in Microsoft Excel and Power BI.

It helps users import, clean, combine, reshape, and prepare data from multiple sources before analysis.

6) What are Filters in Power BI ?

Filters in Power BI are features that allow users to limit, control, and display only the specific data they want to analyze.

Types of Filters in Power BI

- Visual Filter – applies to a single visual (chart or graph)
- Page Filter – applies to all visuals on a page

SECTION – B

II. Answer any four questions. Each question carries five marks. (4x5=20)

7) Write a note on Data Analytics Life Cycle

The Data Analytics Life Cycle is a structured process that describes the different stages involved in collecting, processing, analyzing, and interpreting data in order to make informed decisions.

It provides a systematic approach to solve data-related problems in organizations.

Stages of Data Analytics Life Cycle

1. Data Discovery / Problem Identification

Understanding the business problem, objectives, goals, and requirements.

Identify what decisions need to be made using the data.

2. Data Collection

Collecting data from various sources such as databases, surveys, sensors, social media, or files.

3. Data Preparation / Cleaning

Removing errors, missing values, duplicates, and organizing the data into usable format (ETL: Extract, Transform, Load).

4. Data Analysis / Modeling

Applying statistical methods, algorithms, or machine learning models to uncover patterns, trends, and insights.

5. Data Visualization & Interpretation

Presenting analyzed data using charts, dashboards, and graphs to communicate results clearly.

6. Decision Making

Converting insights into actionable decisions or strategies for solving the business problem.

7. Deployment & Monitoring

Implementing the solution in the real environment and continuously monitoring performance for improvements.

8) Define Hypothesis. Explain the purpose of ANOVA in Hypothesis testing.

A Hypothesis is a proposed statement or assumption about a population parameter that can be tested using statistical methods.

It is made based on sample data to check whether the assumption is true or false.

Purpose of ANOVA in Hypothesis Testing

ANOVA (Analysis of Variance) is a statistical technique used to test whether there is a significant difference between the means of three or more groups.

It helps determine if observed differences among group means are due to real effects or just random chance.

Why ANOVA is Used

- To compare more than two group means at the same time
- To test if all groups have the same average value
- To avoid multiple t-tests and reduce error probability
- Used in experiments involving multiple categories or treatments

- Example
- Comparing the average performance of students in three different teaching methods to see if at least one teaching method is better than the others.

9) What are the various steps involved in any Analytics Project ?

An Analytics Project follows a structured sequence of steps to collect, process, analyze, and interpret data to solve business problems. These steps ensure that data-driven decisions are accurate and useful.

Steps Involved in an Analytics Project

1. Problem Definition / Requirement Understanding

Identify the business problem, project goals, and what insights or decisions are needed.

2. Data Collection

Gather relevant data from different sources such as databases, surveys, sensors, files, or online sources.

3. Data Cleaning / Preparation

Remove errors, missing values, duplicates, and format the data for analysis (ETL: Extract, Transform, Load).

4. Data Exploration / Analysis

Use statistical techniques, analytical methods, or ML models to identify patterns and relationships.

5. Modeling & Evaluation

Apply analytical models or algorithms and test their performance using evaluation metrics.

6. Data Visualization & Interpretation

Present the results using charts, dashboards, or reports to understand insights clearly.

7. Decision Making / Insights Delivery

Use findings to support decisions, recommend actions, or create strategies.

8. Deployment & Monitoring

Implement the analytical solution in real environment and monitor performance for improvement.

10) State and prove Baye's Theorem.

Bayes' theorem is used to determine the conditional probability of event A when event B has already occurred.

Statement of Bayes' Theorem

Bayes' Theorem states that the probability of an event B occurring given that another event A has already occurred is equal to the probability of A occurring given B, multiplied by the probability of B, divided by the total probability of A.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Where,

- **P(A)** and **P(B)** are the probabilities of events A and B; also, P(B) is never equal to zero.
- **P(A|B)** is the probability of event A when event B happens,
- **P(B|A)** is the probability of event B when A happens.

- **Steps to Prove Bayes' Theorem**

- **Step 1: Definition of Conditional Probability**

- $P(A|B) = P(A \cap B) / P(B)$

From this, we get:

$$P(A \cap B) = P(A|B) \cdot P(B)$$

- **Step 2: Another form of Conditional Probability**

$$P(B|A) = P(A \cap B) / P(A)$$

- **Step 3: Substitute the value of P(A ∩ B) from step 1**

$$P(B|A) = P(A|B) \cdot P(B) / P(A)$$

Hence, Bayes' Theorem is proved

$$P(B|A) = P(A|B) P(B) / P(A)$$

12) What are the advantages of Power BI?

Easy to Use: Power BI provides a simple and user-friendly interface that allows users to create reports and dashboards without programming skills.

Interactive Dashboards: It offers interactive visualizations, filters, and drill-down features that help in better data analysis.

Handles Large Data: Power BI can process and analyze large datasets faster using powerful data models.

Multiple Data Source Connectivity: It connects to various data sources such as Excel, SQL Server, cloud services, web data, etc.

Real-Time Data Updates: Dashboards update automatically when the data source changes.

Integration with Microsoft Tools: Works well with Excel, Power Query, Power Apps, Azure, and Microsoft Teams.

Cost Effective: Power BI provides a free desktop version and affordable paid plans.

Data Security: It offers strong security features like row-level security and data encryption.

SECTION – C

III. Answer any four questions. Each question carries eight marks. (4x8=32)

13) With an example explain the different types of analytics.

1. Descriptive Analytics (What happened?)

This is the most fundamental and common type of analytics. It involves summarizing and describing historical data to understand the current state or past performance.

- **Goal:** To gain a basic understanding of what occurred.
- **Techniques:** Data aggregation, data mining, reporting, visualizations (dashboards, charts, graphs).
- **Example:**
 - **Business:** Generating a monthly sales report that shows the total revenue, number of units sold, and the top-selling products for the last month.
 - **Healthcare:** Calculating the average number of patient readmissions in the past quarter.

2. Diagnostic Analytics (Why did it happen?)

Once you know *what* happened, diagnostic analytics helps you delve deeper to uncover the root causes of those outcomes. It focuses on identifying anomalies and determining the factors that contributed to a specific result.

- **Goal:** To understand the reasons and drivers behind a particular event or trend.
- **Techniques:** Drill-down, data discovery, data mining, correlation analysis, root cause analysis.
- **Example:**
 - **Business:** Analyzing a sudden drop in website traffic by looking at factors like a recent website update, a change in a search engine algorithm, or a drop in a marketing campaign's budget to pinpoint the exact cause.
 - **Manufacturing:** Investigating why the defect rate increased last week by correlating it with factors like raw material batch, machine maintenance logs, or a change in shift workers.

3. Predictive Analytics (What will happen?)

This type of analytics uses historical data and statistical models to forecast future probabilities, trends, and outcomes. It moves from retrospective analysis (looking back) to prospective analysis (looking forward).

- **Goal:** To forecast future events or behavior with a level of statistical confidence.
- **Techniques:** Regression analysis, time series forecasting, machine learning models (e.g., neural networks), and statistical modeling.
- **Example:**
 - Finance: Using a customer's credit history, income, and debt-to-income ratio to predict the probability of defaulting on a loan (Credit Risk Assessment).
 - E-commerce: Analyzing a customer's past purchases and browsing history to recommend products they are likely to buy next.

4. Prescriptive Analytics (What should we do?)

This is the most advanced form of analytics. It builds upon the insights from all three previous types to recommend a specific, optimal course of action to achieve a desired goal or prevent a problem. It doesn't just predict what will happen; it tells you what to do about it.

- **Goal:** To guide decision-making and optimize business processes by recommending actionable steps.
- **Techniques:** Optimization algorithms, simulation, decision trees, and machine learning.
- **Example:**
 - Logistics: Recommending the optimal delivery routes for a fleet of vehicles in real-time by considering traffic, weather, and delivery priority to minimize travel time and fuel cost.
 - Retail: Dynamically adjusting the price of a product based on predicted demand, current inventory levels, and competitor pricing to maximize profit.

14) With a case study explain how analytics has helped the food industry to improve their business.

Case Study: Big Basket's Use of Analytics in the Food Industry

Big data analytics has become a crucial tool for businesses in the food industry, moving beyond simple sales tracking to optimize complex operations and enhance the customer

experience. A notable case study is Big Basket, a major Indian online grocery and food delivery platform, which leverages analytics to significantly improve its business.

1. The Challenge

Big Basket deals with a massive portfolio of perishable goods and non-perishable items, catering to a large, diverse, and geographically spread customer base. The key business challenges include:

- **Inventory and Supply Chain Management:** Predicting demand for thousands of SKUs (Stock Keeping Units), especially perishable fresh produce, to minimize spoilage (food loss) and avoid stock-outs.
- **Customer Engagement:** Providing a highly personalized shopping experience to improve customer retention and increase order frequency.
- **Logistics:** Optimizing delivery routes and fulfilling orders efficiently.

2. Analytics Implementation (The Solution)

Big Basket developed and implemented advanced machine learning and big data models to tackle these challenges:

A. Demand and Inventory Forecasting (Predictive Analytics)

- **Model:** They use machine learning models that analyze extensive historical data (sales trends, seasonality, festival demand, promotions) combined with real-time data to forecast demand at a very granular level.
- **Impact:** By accurately predicting demand, they can:
 - **Reduce Spoilage/Food Waste:** Ordering precisely the required amount of perishable inventory, significantly reducing food loss in their warehouses and distribution centers.
 - **Optimize Stocking:** Ensuring popular items are always in stock, improving the availability of products and reducing customer disappointment.

B. Personalized Customer Experience (Customer Analytics)

- **Tool:** Big Basket implemented a 'Smart Basket' feature, among others, powered by historical purchase data analysis.
- **Impact:**
 - **Faster Checkout:** The system analyzes a customer's past purchases and frequency to populate a "smart basket" with items the customer is likely to buy again (e.g., reminding them to add milk or a specific brand of atta). This

reduces search time and makes the shopping process faster and more convenient.

- Targeted Marketing: Analyzing consumer behavior and purchasing patterns helps segment customers. This allows for highly personalized recommendations, promotions, and marketing campaigns, increasing sales and customer loyalty.

C. Operational Efficiency (Prescriptive and Diagnostic Analytics)

- Model: They use analytics to optimize their logistics, including warehouse layout, picking paths, and last-mile delivery routes.
- Impact:
 - Reduced Delivery Time: Optimizing delivery routes based on real-time traffic and order density ensures faster delivery times, a critical factor for customer satisfaction in the e-commerce food space.
 - Cost Savings: Efficient route planning and optimized inventory management directly reduce operational costs, improving the profit margin in a high-volume, low-margin business.

3. Conclusion

The Big Basket case demonstrates how analytics transforms the food business from a reactive to a proactive model. By leveraging big data to predict consumer demand, personalize the shopping experience, and optimize their complex supply chain, Big Basket was able to effectively reduce operational costs (especially waste) and significantly boost customer satisfaction and retention, securing a leading position in the highly competitive Indian online grocery market.

16) a) What are the various types of refresh options provided in power BI ? (3+5)

b) What are the building blocks of Microsoft Power BI ? Explain.

a) Various Types of Refresh Options in Power BI

In Power BI, refreshing the data means updating the semantic model (dataset) with the latest information from the original data sources. The various types of refresh options available, primarily in the Power BI Service, include:

Refresh Type	Description	Primary Use Case
1. Manual Refresh / On-Demand Refresh	The user (owner or admin) manually triggers	Used when new data is added, and the user needs to view the

Refresh Type	Description	Primary Use Case
	the refresh operation by clicking the "Refresh now" button in the Power BI service or the "Refresh" button in Power BI Desktop.	changes immediately, or for ad-hoc updates.
2. Scheduled Refresh	The data refresh is configured to run automatically at pre-set times and frequency (e.g., daily, multiple times a day). This is the most common method for Import mode models.	Used for regularly updating reports and dashboards to ensure data accuracy without manual intervention.
3. Incremental Refresh	An advanced feature where only the <i>latest</i> data changes (new or updated rows) are loaded, leaving historical data partitions untouched. This significantly speeds up the refresh process.	Used for very large semantic models (datasets) to improve refresh performance and reduce resource consumption.
4. Automatic Page Refresh (for Direct Query)	A report-level setting that automatically refreshes all visuals on a page based on a configured interval or when data changes are detected at the source (Change Detection).	Used for reports that require near-real-time monitoring of live data sources (when using Direct Query mode).

Refresh Type	Description	Primary Use Case
5. OneDrive Refresh (File Synchronization)	This synchronizes the Power BI file itself (e.g., a Power BI Desktop file saved to OneDrive) with the Power BI Service, but it does not reload the data from the original source.	Used to ensure the Power BI file structure and metadata are up-to-date in the Power BI Service. A data refresh (Manual/Scheduled) is still required to get new source data.
6. API/Programmatic Refresh	Refresh is triggered via external applications using the Power BI REST APIs, often integrated with automation tools like Power Automate.	Used for custom, event-driven refresh scenarios or for more granular control over the refresh process.

b) Building Blocks of Microsoft Power BI

The Microsoft Power BI ecosystem is built upon five fundamental components, which are often referred to as its "building blocks." They form a logical flow from data connection to insight sharing.

Building Block	Description	Simple Example (Sales Analysis)
1. Dataset / Semantic Model	The prepared, cleaned, and modeled data that Power BI uses. It's the foundation upon which all visuals are built. It includes the data tables, relationships, and calculated measures (DAX).	Example: A single unified dataset created by combining an Excel file (with Sales Transactions) and a SQL Server Table (with Product Information) into one related model.

Building Block	Description	Simple Example (Sales Analysis)
2. Visualization (Visual)	A single graphical representation of data. It is a single chart, graph, map, or card that highlights a specific data point or trend.	Example: A Bar Chart showing "Total Sales Amount by Region" OR a Card Visual displaying the "Total Revenue for the Last Month" (\$500K).
3. Report	A multi-perspective view of a single dataset. It consists of one or more pages, where each page contains multiple related visualizations (visuals), filters, and slicers for detailed analysis.	Example: A three-page Sales Performance Report. Page 1 might show overall Sales Trends, Page 2 might detail Sales by Product Category, and Page 3 might display Sales by Customer Demographics.
4. Tile	A snapshot of a single visualization that has been pinned from a Report to a Dashboard. It is the key building block of a Dashboard.	Example: The Card Visual showing "Total Revenue for the Last Month" (\$500K) is pinned to the Executive Dashboard.
5. Dashboard	A single-page, high-level summary view that aggregates key tiles from one or more reports. It is designed to provide quick, at-a-glance monitoring of Key Performance Indicators (KPIs) and is only available in the Power BI Service.	Example: An Executive Sales Dashboard with three key tiles: 1) Last Month's Revenue (\$500K), 2) Profit Margin Trend (a line chart), and 3) Top 5 Selling Products (a bar chart).

17) a) What is the purpose of COUNT, COUNTA, COUNTBLANK and COUNTIF in Excel?(4+4)

(b) List the difference between Logistic Regression and Linear Regression.

a) Purpose of COUNT, COUNTA, COUNTBLANK, and COUNTIF in Excel

These are essential statistical functions in Microsoft Excel used for aggregation and data quality analysis:

Function	Purpose	Example
COUNT	Counts the number of cells in a range that contain numeric values (numbers, dates, times).	<i>=COUNT (A1:A10): If A1 is text, A2 is 10, and A3 is empty, the result is 1.</i>
COUNTA	<i>Counts the number of cells in a range that are not empty (non-blank). It counts numbers, text, dates, times, logical values (TRUE/FALSE), and error values.</i>	<i>=COUNTA (A1:A10): If A1 is text, A2 is 10, and A3 is empty, the result is 2.</i>
COUNTBLANK	<i>Counts the number of cells in a range that are completely empty (blank).</i>	<i>=COUNTBLANK (A1:A10): If A1 is text, A2 is 10, and A3 is empty, the result is 1.</i>
COUNTIF	<i>Counts the number of cells within a range that meet a single, specific criterion (condition) set by the user.</i>	<i>=COUNTIF (B1:B10, ">50"): Counts how many values in the range B1:B10 are greater than 50.</i>

b) Difference Between Logistic Regression and Linear Regression

18) a) Differentiate between Dashboard and Reports. (4+4)

b) Explain the different visualisation techniques used for spatial data.

a) Differentiate between Dashboard and Reports

In the context of Business Intelligence (BI) tools like Power BI, reports and dashboards have distinct purposes and characteristics

Feature	Dashboard	Report
Purpose	High-Level Monitoring. Provides a quick, at-a-glance summary of Key Performance Indicators (KPIs) and business health.	In-Depth Analysis. Provides detailed, multi-perspective views for data exploration, trend analysis, and answering specific questions.
Length/Pages	Single Page (often called a canvas). Designed to fit on one screen.	Multi-Page (can have multiple tabs or pages). Designed to accommodate extensive detail.
Source Data	Can aggregate visualizations (Tiles) from Multiple Reports and Multiple Datasets to show a unified view.	Generally built on a Single Dataset (Semantic Model).
Interactivity	Limited. Users can click a "Tile" to navigate to the underlying report, but usually cannot apply dynamic filters (like Slicers) directly on the dashboard.	Highly Interactive. Allows for complex filtering, sorting, cross-filtering, and drilling down into granular data.
Creation Tool	Created only in the Power BI Service (online).	Created primarily in Power BI Desktop and then published to the Service.
Data Focus	Often focuses on Real-time or near real-time data using "live tiles" for instant monitoring.	Typically focuses on Historical data and deep-dive comparisons, requiring periodic scheduled refresh.
Audience	Executives, Managers, and Decision-Makers who need a quick summary.	Analysts, Power Users, and Teams needing to explore detailed data.

b) Different Visualization Techniques for Spatial Data

Spatial data (or Geospatial data) includes information about the location and distribution of objects or events on the Earth's surface. Effective visualization uses maps to highlight patterns, clusters, and relationships

1. Choropleth Map (Thematic Map)

Technique	Description	Example
Choropleth Map	Geographic areas (like states, counties, or districts) are shaded or colored based on a specific data value. The color intensity represents the magnitude of the variable. Best used for standardized data (rates, ratios, percentages).	Example: A map of India where each state is colored with a shade of blue based on its Literacy Rate . Darker blue states have a higher literacy rate, and lighter blue states have a lower rate.

2.Point Maps / Dot Density Maps

Technique	Description	Example
Point Map	Uses individual markers (dots) plotted on a map using exact latitude and longitude coordinates to show the precise location of events or entities.	Example: A map of a city showing the location of every reported traffic accident last month, with each accident represented by a small red dot.
Dot Density Map	Uses dots to represent a fixed number of items to show concentration. If an area has a high concentration, the dots will be dense.	Example: A map of a country where one dot represents 1,000 people . Areas with higher population density will have a heavy cluster of dots, appearing darker.

3. Proportional Symbol / Bubble Maps

Technique	Description	Example
Proportional Symbol Map	Points are plotted on the map, and the size of the symbol (circle or bubble) is scaled proportionally to the value of a specific data variable at that location. The size immediately indicates the magnitude.	Example: A map of major world cities where the size of the bubble over each city is proportional to the city's total population. The bubble over Tokyo would be much larger than the bubble over a smaller city.

4. Heat Maps (Density Maps)

Technique	Description	Example
Heat Map	Uses a gradient of colors (like a thermal image, from blue/cool to red/hot) over a geographic area to show the concentration or density of a large volume of data points, regardless of administrative boundaries.	Example: A map of a retail location showing customer activity. Areas in bright red indicate high foot traffic or where customers spend the most time, while blue areas indicate low activity.

5. Flow Maps / Connector Maps

Technique	Description	Example
Flow Map	Uses lines or arrows to illustrate movement, connection, or interaction between two or more locations. The thickness or color of the line often represents the volume or magnitude of the flow.	Example: A map showing airline traffic between major global airports. The line connecting New York to London would be much thicker than the line connecting a smaller regional airport, indicating a higher volume of travelers.